

## ***B.Tech. Degree IV Semester Examination April 2013***

### **ME 406 MANUFACTURING PROCESS (2006 Scheme)**

Time : 3 Hours

Maximum Marks : 100

#### **PART A (Answer *ALL* questions)**

(8 × 5 = 40)

- I. (a) Define the term casting.  
(b) Explain the significance of gating ratio.  
(c) List out the various casting defects.  
(d) Explain shell moulding.  
(e) Describe the principle of rolling.  
(f) Explain embossing and coining.  
(g) Describe the principle of resistance welding.  
(h) Explain the applications of thermit welding.

#### **PART B**

(4 × 15 = 60)

- II. (a) Discuss the typical applications of metal casting process.  
(b) Explain the desirable properties of moulding sand.

**OR**

- III. (a) Describe the different types of cores.  
(b) Write a brief note on the type of patterns and colour codes.

- IV. Explain with neat sketches the procedure involved in centrifugal casting.

**OR**

- V. Write notes on: (i) Die casting  
(ii) Casting cleaning  
(iii) Design for eliminating defects in casting

- VI. (a) Distinguish between hot working and cold working.  
(b) Explain smith forging.

**OR**

- VII. (a) What are the different types of defects found in forging?  
(b) Explain drawing, shearing and spinning operation in sheet metal.

- VIII. (a) With a neat sketch explain the characteristics of the different types of flames used in oxy-acetylene welding.  
(b) Explain gas cutting.

**OR**

- IX. Write notes on:  
(i) Soldering  
(ii) Brasing  
(iii) Laser beam welding.

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